Question 1:
The user enters 123
Which of the following code snippets will print 124 to the monitor? Choose three.

A.	1 | num = int (input ('Please enter your number: ')) 2 | print (num + 1)
B.	1 | num = input ('Please enter your number: ') 2 | print (num + 1)
C.	1 | num = eval (input ('Please enter your number: ')) 2 | print (num + 1)
D.	1 | num = input ('Please enter your number: ') 2 | print (int (num) + 1)

Question 2:
What is the expected output of the following code?

```
x = (1, 4, 7, 9, 10, 11)
y = {2: 'A', 4: 'B', 6: 'C', 8: 'D', 10: 'E', 12: 'F'}
res = 1
for z in x:
    if z in y:
        res += z
print(res)
```

A. 14
B. 6
C. 15
D. 22
E. None of the above.

Question 3:
Evaluate the following Python arithmetic expression:

```
(3 * (1 + 2) ** 2 - (2 ** 2) * 3)
```

What is the result?

A. 15
B. 3
C. 69
D. 13

Question 4:
Consider the following list.

```
data = [1, 5, 10, 19, 55, 30, 55, 99]
```

Which of the code snippets below would produce a new list like the following?

```
[1, 5, 10, 99]
```

A.	1 | data.remove(5) 2 | data.remove(19) 3 | data.remove(55)
B.	1 | data.pop(5) 2 | data.remove(19) 3 | data.remove(55) 4 | data.remove(55)
C.	1 | data.pop(1) 2 | data.pop(3) 3 | data.pop(4) 4 | data.pop(6)
D.	1 | data.pop(5) 2 | data.pop(19) 3 | data.pop(55)
E.	None of the above.

Question 5:
You are writing a Python program that evaluates an arithmetic formular. The formular is described as b equals a multiplied by negative one, then raised to the second power,

where a is the value that will be input and b is the result.

```
a = eval(input('Enter a number for the equation: '))
```

Which of the following is a valid expression for the given requirement?

A. b = (a-) ** 2
B. b = (a) ** -2
C. b = -(a) ** 2
D. b = (-a) ** 2

Question 6:
What is the default value of encoding in the string function encode()?

A. ascii
B. utf-8
C. qwerty
D. utf-16

Question 7:
What is the expected output of the following code?

```
def get_names():
    names = ['Peter', 'Paul', 'Mary', 'Jane', 'Steve']
    return names[2:]

def update_names(names):
    res = []
    for name in names:
        res.append(name[:3].upper())
    return res

print(update_names(get_names()))
```

A. [‘JA’ , ‘ST’]
B. [‘JAN’ , ‘STE’]
C. [‘MA’ , ‘JA’ , ‘ST’]
D. [‘MAR’ , ‘JAN’ , ‘STE’]

Question 8:
What is the expected output of the following code?

```
list1 = ['Peter', 'Paul', 'Mary', 'Jane']
list2 = ['Peter', 'Paul', 'Mary', 'Jane']

print(list1 is not list2)
print(list1 != list2)

list1 = list2

print(list1 is not list2)
print(list1 != list2)
```

A.	1 | True 2 | True 3 | False 4 | False
B.	1 | True 2 | False 3 | False 4 | True
C.	1 | True 2 | False 3 | False 4 | False
D.	1 | True 2 | False 3 | True 4 | False

Question 9:
What is the expected output of the following code?

```
try:
    raise Exception
except BaseException:
    print('1')
except Exception:
    print('2')
except:
    print('3')
```

A. 3
B. 1
C. 2
D. The code is erroneous.

Question 10:
You want to write Hello world to the file data.txt
Which of the following code snippets do you need?

A.	1 | f = open (‘data.txt’) 2 | f.write (‘Hello world’) 3 | f.close ( )
B.	1 | f = open (‘data.txt’ , ‘b’) 2 | f.write (‘Hello world’) 3 | f.close ( )
C.	1 | f = open (‘data.txt’ ‘r’) 2 | f.write (‘Hello world’) 3 | f.close ( )
D.	1 | f = open (‘data.txt’ ‘w’) 2 | f.write (‘Hello world’) 3 | f.close ( )

Question 11:
What is the expected output of the following code?

```
1  print(type(+1E10))
2  print(type(5.0))
3  print(type('True'))
4  print(type(False))
```

A.	1 | <class ‘float’ > 2 | <class ‘float’ > 3 | <class ‘bool’ > 4 | <class ‘bool’ >
B.	1 | <class ‘float’ > 2 | <class ‘float’ > 3 | <class ‘str ‘ > 4 | <class ‘bool’ >
C.	1 | <class ‘int’ > 2 | <class ‘float’ > 3 | <class ‘str ‘ > 4 | <class ‘bool’ >
D.	1 | <class ‘int’ > 2 | <class ‘float’ > 3 | <class ‘bool’ > 4 | <class ‘bool’ >

Question 12:
Consider the following code.

```
import random
print(int(random.random() * 7))
```

Which of the following statements is correct?

A. It will print 7
B. It will print a random integer value from 0 to 6
C. It will print a random integer value from 0 to 7
D. It will print a random integer value from 1. to 7

Question 13:
Consider the following code.

```
def get_data(filename, mode):
    import os
    if os.path.isfile(filename):
        with open(filename, mode) as file:
            return file.readline()
    else:
        return
```

Which of the statements about this code are true? Choose Two.

A. This function returns None if the file does not exist.
B. This function returns the first line of the file if it is available.
C. This function returns the last line of the file if it is available.
D. This function returns the total data that is present in the file.

Question 14:

What is the expected output of the following code?

```
box = {}
jars = {}
crates = {}

box['biscuit'] = 1
box['cake'] = 3

jars['jam'] = 4

crates['box'] = box
crates['jars'] = jars

print(len(crates[box]))
```

A. 3
B. 2
C. 4
D. 1
E. The code is erroneous.

Question 15:
What does the following code do?

```
def a(b, c, d):
    pass
```

A. Defines an empty class.
B. Defines a function, which does nothing.
C. Defines a list and initialises it.
D. Defines a function, which passes its parameters through.
E. None of the above.

Question 16:
What is the expected output of the following code?

```
file = open('data.txt', 'r')
txt = "I'm gonna make him an offer he can't refuse."
file.writelines(txt)
file.seek(0)
lines = file.readlines()
print(lines)
file.close()
```

A. [“I’m gonna make him an offer he can’t refuse.”]
B. None
C. [ ]
D. The code is erroneous.

Question 17:
Which of the approachable exceptions: branches is taken into consideration when an exception occurs?

A. The last matching branch.
B. Any of the matching branches.
C. The first matching branch.
D. None of the above.

Question 18:
What is the expected output of the following code?

```
data = ['peter', 'paul', 'mary']
for d in data:
    data.append(d.upper())
print(data)
```

A. [‘Peter’ , ‘Paul’ , ‘Mary’]
B. [‘PETER’ , ‘PAUL’ , ‘MARY’]
C. The code will enter an infinite loop.
D. The code is erroneous.

Question 19:
What is the expected output of the following code?

```
def func1(a):
    return a ** a

def func2(a):
    return func1(a) * func1(a)

print(func2(2))
```

A. 2
B. 16
C. 4
D. The code is erroneous.

Question 20:
The module has been imported using the following line.

```
from mod import func
```

How can you invoke the function?

A. func ( )
B. mod : func ( )
C. mod : : func ( )
D. mod.func ( )

Question 21:
What is the expected output of the following code?

```
def func1(x):
    return str(x)

def func2(x):
    return str(2 * x)

print(func1(1) + func2(2))
```

A. The code is erroneous.
B. 14
C. 3
D. 5

Question 22:
An operator able to check whether two values are equal, is coded as:

A. ==

B. =
C. ===
D. is

Question 23:
You want the name the user puts in to be written back to the monitor. What snippet would you insert in the line indicated below:

```
print('Enter Your Name: ')
# insert your code here
print(name)
```

A. input ('name')
B. name = input
C. name = input ( )
D. input (name)

Question 24:
What is the expected output of the following code if the user enters 3 and 2?

```
x = int(input())
y = int(input())
x = x % y
x = x % y
y = y % x
print(y)
```

A. 0
B. 2
C. 3
D. 1

Question 25:
Which of the code snippets below will print the following to the monitor?

```
Paul
Mary
Jane
```

A.	1 | data = ['Peter' , 'Paul' , 'Mary' , 'Jane'] 2 | for d in data: 3 | if len(d) == 4 4 | print (d)
B.	1 | data = ['Peter' , 'Paul' , 'Mary' , 'Jane'] 2 | for d in data: 3 | if len(d) != 4: 4 | print(d)
C.	1 | data = ['Peter' , 'Paul' , 'Mary' , 'Jane'] 2 | for d in data: 3 | print (d)
D.	1 | data = ['Peter' , 'Paul' , 'Mary' , 'Jane'] 2 | da = data[1:] 3 | for d in data: 4 | print(d)

Question 26:
What is the expected output of the following code?

```
data = {}
data[1] = 1
data['1'] = 2
data[1.0] = 4

res = 0
for d in data:
    res += data[d]

print(res)
```

A. The code is erroneous.
B. 7
C. 6
D. 3

Question 27:
What is the expected output of the following code?

```
data1 = 'a', 'b'
data2 = ('a', 'b')
print(data1 == data2)
```

A. False
B. 0
C. 1
D. True

Question 28:
Which of the literals below is not a valid function name?

A. _function1
B. Function_1
C. Func_1_tion
D. Function1
E. 1Function

Question 29:
The pyc file contains ...

A. Python source code.
B. a Python interpreter.
C. compiled Python bytecode.
D. a Python compiler.

Question 30:
Consider the following code.

```
x = float('23.42')
```

Which of the following expressions will evaluate to 2?

A. int (x) + False
B. bool (x)
C. str (x)
D. bool (x) + True

Question 31:
An assertion can be used to:

A. Import a module.
B. Stop the program when some data have improper values.
C. Make the Programmer more assertive.
D. None of the above.

Question 32:
What is the expected output of the following code?

```
print(1 // 2)
```

A. 0.0
B. 0.5
C. 0
D. None of the above.

Question 33:
Which of the following operators can be used with strings?

1) +
2) *
3) -
4) in

A. 1, 2, 3
B. 1, 2, 4
C. 1, 2
D. 1, 2, 3, 4

Question 34:
What is the expected output of the following code?

```
class A:
    def __init__(self):
        self.text = 'abc'
        self.count = 0

    def __iter__(self):
        return self

    def __next__(self):
        if self.count == len(self.text):
            raise StopIteration
        value = self.text[self.count]
        self.count += 1
        return value

for x in A():
    print(x, end='')
```

A. 210
B. abc
C. 012
D. cba

Question 35:
What is the expected output of the following code?

```
print(not 0)
print(not 23)
print(not '')
print(not 'Peter')
print(not None)
```

A.	1 | True 2 | False 3 | False 4 | False 5 | True
B.	1 | False 2 | False 3 | True 4 | False 5 | True
C	1 | True 2 | False 3 | True 4 | False 5 | True
D.	1 | True 2 | False 3 | True 4 | False 5 | True

Question 36:
Presume f is a file opened in read mode. What will be the type of data after the following line?

```
data = f.readlines()
```

A. dictionary
B. string
C. list
D. tuple

Question 37:
A function returning a list of all entities available in a module is called:

A. listmodule ( )
B. entities ( )
C. dir ( )
D. content ( )

Question 38:
What is the expected output of the following code?

```
data = {'z': 23, 'x': 7, 'y': 42}

for _ in sorted(data):
    print(data[_], end=' ')
```

A. 7 42 23
B. 42 23 7
C. The code is erroneous.
D. 7 23 42

Question 39:
Which one of the following is true?

A. packages can contain modules
B. modules can contain modules
C. modules can contain packages
D. None of the above.

Question 40:
What is the expected output of the following code?

```
x = [1, 2, 3, 4, 5, 6, 7, 8, 9]
x[::2] = 10, 20, 30, 40, 50, 60
print(x)
```

A. [10, 2, 20, 4, 30, 6, 40, 8, 50, 60]
B. [1, 2, 10, 20, 30, 40, 50, 60]
C. The code is erroneous.
D. [1, 10, 3, 20, 5, 30, 7, 40, 9, 50, 60]

Question 41:
What is the expected output of the following code?

```
data = 'Hello@Peter!!'
print(data.lower())
```

A. None
B. hello@Peter! !
C. hellopeter
D. hello@peter! !

Question 42:
What is the expected output of the following code?

```
print(chr(ord('p') + 3))
```

A. q
B. t
C. r
D. s

Question 43:
What is the expected output of the following code?

```
try:
    print('try')
    print(10 / 0)
except:
    print('except')
else:
    print('else')
finally:
    print('finally')
```

A.	1 | try 2 | except 3 | else 4 | finally
B.	1 | try 2 | except 3 | finally
C.	1 | try 2 | else 3 | finally
D.	1 | try 2 | finally
E.	1 | except 2 | finally

Question 44:
A method able to read data from a file into a bytes object, is named:

A. readinto( )
B. readout ( )
C. readin( )
D. read( )

Question 45:
This expression is:

```
'mike' > 'Mike'
```

A. false
B. true
C. erroneous
D. None of the above.

Question 46:
Analyse the following code fragments that assigns a boolean value to the variable even?

```
num = 42

# Code-1
if num % 2 == 0:
    even = True
else:
    even = False

# Code-2
even = True if num % 2 == 0 else False

# Code-3
even = num % 2 == 0
```

A. Code-3 has a syntax error because you attempt to assign a number to even
B. Code -2 has a syntax error because you cannot have True and False literals in the conditional expression.
C. All three are correct, but Code-1 is preferred.
D. All three are correct, but Code-2 is preferred.
E. All three are correct, but Code-3 is preferred.

Question 47:
Consider the following code.

```
x = [0, 1, 2]
x[0], x[2] = x[2], x[0]
```

What does the second assignment do?

A. It extends the list.
B. It doesn't change the list.
C. It reverses the list.
D. It shortens the list.

Question 48:
How much will the delivery cost be, if the order value is 1700 and the state is FL (Florida)?

```python
order = int(input('Please enter the order value: '))
state = input('Please enter the state (as postal abbreviation): ')
delivery = 0

if state in ['NC', 'SC', 'VA']:
    if order <= 1000:
        delivery = 70
    elif 1000 < order < 2000:
        delivery = 80
    else:
        delivery = 90
else:
    delivery = 50
if state in ['GA', 'WV', 'FL']:
    if order > 1000:
        delivery += 30
    if order < 2000 and state in ['WV', 'FL']:
        delivery += 40
    else:
        delivery += 25
print(delivery)
```

A. 90
B. 80
C. 105
D. 120

Question 49:
You are creating a Python script to evaluate input and check for upper and lower case.
Code segment 1:

```python
else:
    print(name, 'is mixed case.')
```

Code segment 2:

```
else:
    print(name, 'is lower case.')
```

Code segment 3:

```
name = input('Enter your name: ')
```

Code segment 4:

```
else:
    print(name, 'is upper case.')
```

Code segment 5:

```
elif name.upper() == name:
    print(name, 'is all upper case.')
```

Code segment 6:

```
if name.lower() == name:
    print(name, 'is all lower case.')
```

Which four code segments should you use to develop the solution?

A.	Code segment 1 Code segment 3 Code segment 5 Code segment 6
B.	Code segment 3 Code segment 6 Code segment 5 Code segment 1
C.	Code segment 3 Code segment 6 Code segment 5 Code segment 2
D.	Code segment 3 Code segment 6 Code segment 5 Code segment 4

Question 50:
What is the expected output of the following code?

```
x = 0
y = 1
x = x ^ y
y = x ^ y
y = x ^ y
print(x, y)
```

A. 1 0
B. 1 1
C. 0 1
D. 0 0

Question 51:
Consider the file peoples.txt:

```
Peter:10
Paul:20
Mary:30
Jane:40
```

What is the expected output of the following code? (The code is in a file in the same directory than peoples.txt)

```
points = 0
try:
    file = open('peoples.txt', 'r')
    data = file.readlines()
    for d in data:
        points += float(d.split(':')[1])
    file.close()
    print(points)
except:
    print('The file could not be opened!')
```

A. 10.0
B. The file could not be opened.
C. 100.0
D. 100

Question 52:
What is the expected output of the following code?

```
def func1(param):
    return param

def func2(param):
    return param * 2

def func3(param):
    return param + 3

print(func1(func2(func3(1))))
```

A. 8
B. 1
C. 6
D. 3

Question 53:
What is the expected output of the following code?

```
a = 1
b = 0
c = a & b
d = a | b
e = a ^ b
print(c + d + e)
```

A. 3
B. 1
C. 2
D. 0

Question 54:
What would you insert instead ??? so that the program prints TRUE to the monitor?

```
w = 7
x = 3
y = 4
z = True
a = w + x * y
b = w + x / z

if ???:
    print('TRUE')
else:
    print('FALSE')
```

A. a < b
B. a > b
C. a == b
D. a <= b

Question 55:
What is the expected output of the following code?

```
data = [[42, 17, 23, 13], [11, 9, 3, 7]]
res = data[0][0]
for da in data:
    for d in da:
        if res > d:
            res = d
print(res)
```

A. 13
B. The code is erroneous.
C. 42
D. 3

Question 56:
What is the expected output of the following code?

```
x = {(1, 2): 1, (2, 3): 2}
print(x[1, 2])
```

A. 1
B. { (2, 3): 2}
C. The code is erroneous.
D. { (1, 2): 1}

Question 57:
Consider the following code.

```
room = input('Enter the room number: ')
rooms = {101: 'Gathering Place', 102: 'Meeting Room'}
if not room in rooms:
    print('The room doesn\'t exist.')
else:
    print('The room name is: ' + rooms[room])
```

Why is it not working?

A. Misnamed variable(s)
B. Invalid Syntax
C. Mismatched data type(s)
D. None of the above.